Week 1

Ubuntu login : pranav pranav@12

Wazuh login : admin O.2tX3wTUOAf7d2V+rj5bv6DRWN0UeIU

Objective

To design and implement a basic **Security Operations Center (SOC) infrastructure** by deploying the Wazuh Manager on a central server and integrating endpoint systems using Wazuh agents. The objective is to establish secure communication between the server and agents for continuous log collection and monitoring

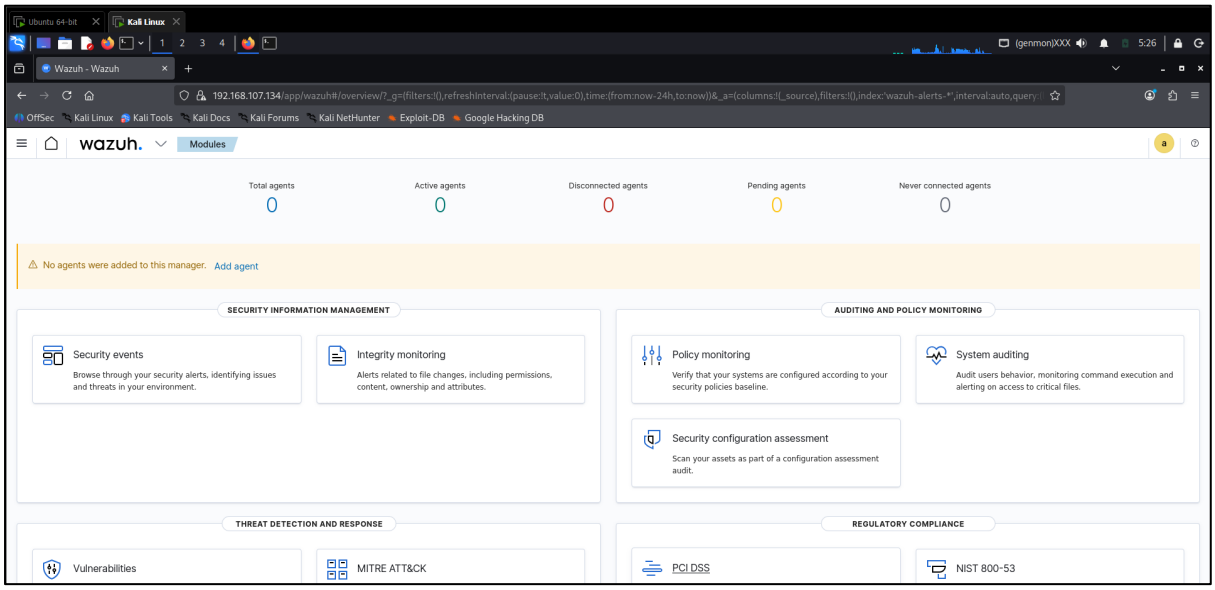
Installing wazuh on ubuntu server

```
pranav@project2:~$ ping -c 4 google.com
PING google.com (142.251.42.238) 56(84) bytes of data.
64 bytes from pnbomb-aw-in-f14.1e100.net (142.251.42.238): icmp_seq=1 ttl=128 time=21.5 ms
64 bytes from pnbomb-aw-in-f14.1e100.net (142.251.42.238): icmp_seq=2 ttl=128 time=8.78 ms
64 bytes from pnbomb-aw-in-f14.1e100.net (142.251.42.238): icmp_seq=3 ttl=128 time=8.65 ms
64 bytes from pnbomb-aw-in-f14.1e100.net (142.251.42.238): icmp_seq=4 ttl=128 time=10.7 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 8.645/12.409/21.476/5.299 ms
pranav@project2:~$ curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh
pranav@project2:~$ ls
wazuh-install.sh
pranav@project2:~$ sudo bash wazuh-install.sh -a
22/03/2026 08:01:21 INFO: Starting Wazuh installation assistant. Wazuh version: 4.7.5
22/03/2026 08:01:21 INFO: Verbose logging redirected to /var/log/wazuh-install.log
22/03/2026 08:01:21 ERROR: The recommended systems are: Red Hat Enterprise Linux 7, 8, 9; CentOS 7, 8; Amazon Linux 2; Ubuntu 16.04, 18.04, 20.04, 22.04. The cu
rrent system does not match this list. Use -i|--ignore-check to skip this check.
```

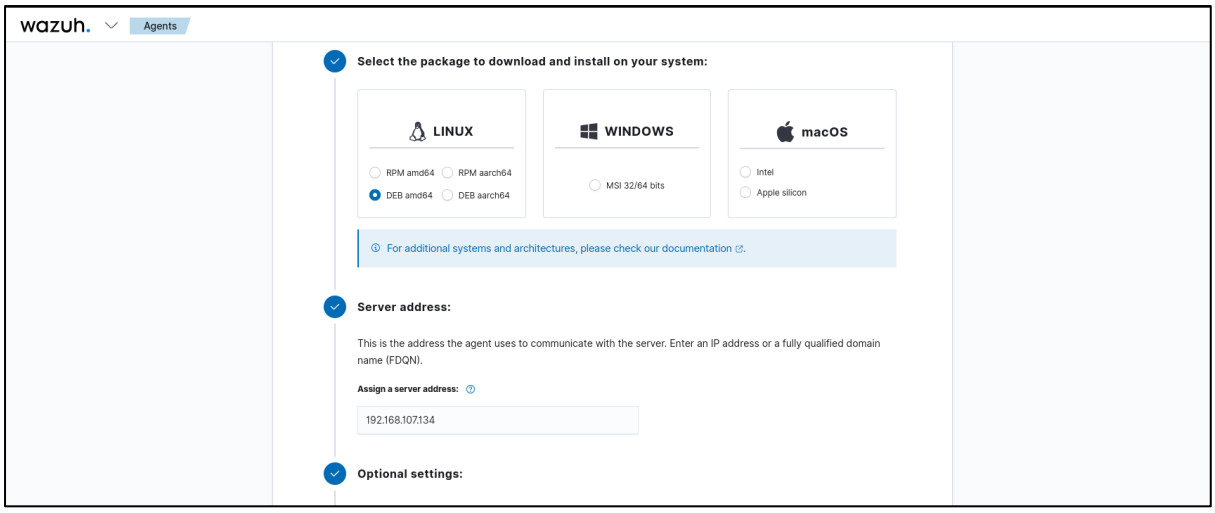
```
pranav@project2:~$ sudo bash wazuh-install.sh -a -i
22/03/2026 09:00:42 INFO: Starting Wazuh installation assistant. Wazuh version: 4.7.5
22/03/2026 09:00:42 INFO: Verbose logging redirected to /var/log/wazuh-install.log
22/03/2026 09:00:52 WARNING: Hardware and system checks ignored.
22/03/2026 09:00:52 INFO: Wazuh web interface port will be 443.
22/03/2026 09:01:03 INFO: Wazuh repository added.
22/03/2026 09:01:03 INFO: --- Configuration files ---
22/03/2026 09:01:03 INFO: Generating configuration files.
22/03/2026 09:01:05 INFO: Created wazuh-install-files.tar. It contains the Wazuh cluster key, certificates, and passwords necessary for installation.
22/03/2026 09:01:05 INFO: --- Wazuh indexer ---
22/03/2026 09:01:05 INFO: Starting Wazuh indexer installation.
22/03/2026 09:02:01 INFO: Wazuh indexer installation finished.
22/03/2026 09:02:01 INFO: Wazuh indexer post-install configuration finished.
22/03/2026 09:02:01 INFO: Starting service wazuh-indexer.
22/03/2026 09:02:16 INFO: wazuh-indexer service started.
22/03/2026 09:02:16 INFO: Initializing Wazuh indexer cluster security settings.
22/03/2026 09:02:27 INFO: Wazuh indexer cluster initialized.
22/03/2026 09:02:27 INFO: --- Wazuh server ---
22/03/2026 09:02:27 INFO: Starting the Wazuh manager installation.
22/03/2026 09:03:52 INFO: Wazuh manager installation finished.
22/03/2026 09:03:52 INFO: Starting service wazuh-manager.
22/03/2026 09:04:11 INFO: wazuh-manager service started.
22/03/2026 09:04:11 INFO: Starting Filebeat installation.
22/03/2026 09:04:19 INFO: Filebeat installation finished.
22/03/2026 09:04:21 INFO: Filebeat post-install configuration finished.
22/03/2026 09:04:21 INFO: Starting service filebeat.
22/03/2026 09:04:23 INFO: filebeat service started.
22/03/2026 09:04:23 INFO: --- Wazuh dashboard ---
22/03/2026 09:04:23 INFO: Starting Wazuh dashboard installation.
22/03/2026 09:05:29 INFO: Wazuh dashboard installation finished.
22/03/2026 09:05:29 INFO: Wazuh dashboard post-install configuration finished.
22/03/2026 09:05:29 INFO: Starting service wazuh-dashboard.
22/03/2026 09:05:30 INFO: wazuh-dashboard service started.
22/03/2026 09:05:57 INFO: Initializing Wazuh dashboard web application.
22/03/2026 09:05:59 INFO: Wazuh dashboard web application initialized.
22/03/2026 09:05:59 INFO: --- Summary ---
22/03/2026 09:05:59 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
    User: admin
    Password: O.2tX3wTUOAf7d2V+rj5bv6DRWN0UeIU
22/03/2026 09:05:59 INFO: Installation finished.
```

After successful installation we now got credential to access wazuh dashboard

This is wazuh initial dashboard accessed from kali machine



Deploying new agent (kali)

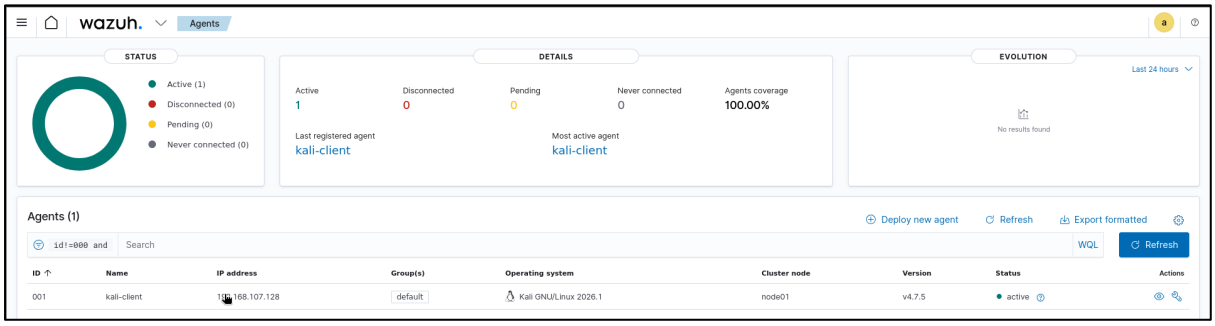


Enabling wazuh agent in Kali terminal

```
┌──(kali㉿kali)-[~]
└─$ wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.5-1_amd64.deb && sudo WAZUH_MANAGER='192.168.107.134' WAZUH_AGENT_NAME='k
ali-client' dpkg -i ./wazuh-agent_4.7.5-1_amd64.deb
--2026-03-22 05:32:48--  https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.5-1_amd64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 108.159.80.123, 108.159.80.89, 108.159.80.93, ...
Connecting to packages.wazuh.com (packages.wazuh.com)|108.159.80.123|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9378818 (8.9M) [application/vnd.debian.binary-package]
Saving to: 'wazuh-agent_4.7.5-1_amd64.deb'

wazuh-agent_4.7.5-1_amd64.deb           100%[===================================================================>]   8.94M  7.67MB/s    in 1.2s

2026-03-22 05:32:49 (7.67 MB/s) - 'wazuh-agent_4.7.5-1_amd64.deb' saved [9378818/9378818]

[sudo] password for kali:
Selecting previously unselected package wazuh-agent.
(Reading database… 445296 files and directories currently installed.)
Preparing to unpack …/wazuh-agent_4.7.5-1_amd64.deb…
Unpacking wazuh-agent (4.7.5-1)…
Setting up wazuh-agent (4.7.5-1)…

┌──(kali㉿kali)-[~]
└─$ sudo systemctl daemon-reload
sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent
Created symlink '/etc/systemd/system/multi-user.target.wants/wazuh-agent.service' → '/usr/lib/systemd/system/wazuh-agent.service'.
```

Verifying agent is running

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl status wazuh-agent
● wazuh-agent.service - Wazuh agent
     Loaded: loaded (/usr/lib/systemd/system/wazuh-agent.service; enabled; preset: disabled)
     Active: active (running) since Sun 2026-03-22 05:35:01 EDT; 1min 14s ago
 Invocation: 15eff56cb886499f8563723cfc7b7421
    Process: 6164 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
      Tasks: 36 (limit: 9277)
     Memory: 1.7G (peak: 1.7G)
        CPU: 57.948s
     CGroup: /system.slice/wazuh-agent.service
             ├─6483 /var/ossec/bin/wazuh-execd
             ├─6494 /var/ossec/bin/wazuh-agentd
             ├─6508 /var/ossec/bin/wazuh-syscheckd
             ├─6521 /var/ossec/bin/wazuh-logcollector
             └─6538 /var/ossec/bin/wazuh-modulesd

Mar 22 05:34:56 kali systemd[1]: Starting wazuh-agent.service - Wazuh agent...
Mar 22 05:34:56 kali env[6164]: Starting Wazuh v4.7.5...
Mar 22 05:34:56 kali env[6164]: Started wazuh-execd...
Mar 22 05:34:57 kali env[6164]: Started wazuh-agentd...
Mar 22 05:34:57 kali env[6164]: Started wazuh-syscheckd...
Mar 22 05:34:58 kali env[6164]: Started wazuh-logcollector...
Mar 22 05:34:59 kali env[6164]: Started wazuh-modulesd...
Mar 22 05:35:01 kali env[6164]: Completed.
Mar 22 05:35:01 kali systemd[1]: Started wazuh-agent.service - Wazuh agent.
```



Conclusion

The Wazuh Manager was successfully installed on the Ubuntu server, and the Kali Linux system was configured as an agent. The agent was able to communicate with the manager and appeared as **Active** in the Wazuh dashboard, confirming successful deployment. This setup established a centralized monitoring environment capable of collecting and analyzing security events in real time.

Week 2

Objective

To implement **File Integrity Monitoring (FIM)** using Wazuh by configuring the syscheck module to monitor critical system and user directories. The objective is to detect unauthorized file creation, modification, and deletion events in real time.

We implemented File Integrity Monitoring using Wazuh, which detects file creation, modification, and deletion events in real time using checksum validation.

```
┌──(kali㉿kali)-[~]
└─$ mkdir ~/testfolder

┌──(kali㉿kali)-[~]
└─$ echo "hello" > ~/testfolder/file.txt

┌──(kali㉿kali)-[~]
└─$ echo "changed" >> ~/testfolder/file.txt

┌──(kali㉿kali)-[~]
└─$ sudo nano /var/ossec/etc/ossec.conf

┌──(kali㉿kali)-[~]
└─$ sudo nano /var/ossec/etc/ossec.conf

┌──(kali㉿kali)-[~]
└─$ sudo nano /var/ossec/etc/ossec.conf

┌──(kali㉿kali)-[~]
└─$ sudo nano /var/ossec/etc/ossec.conf

┌──(kali㉿kali)-[~]
└─$ sudo systemctl restart wazuh-agent

┌──(kali㉿kali)-[~]
└─$ echo "hello" >> /home/kali/test.txt
```

Before:

```
<!-- Frequency that syscheck is executed default every 12 hours -->
<frequency>43200</frequency>

<scan_on_start>yes</scan_on_start>

<!-- Directories to check  (perform all possible verifications) -->
<directories>/etc,/usr/bin,/usr/sbin</directories>
<directories>/bin,/sbin,/boot</directories>
```

After:

```
<!-- Frequency that syscheck is executed default every 12 hours -->
<frequency>60</frequency>

<scan_on_start>yes</scan_on_start>

<!-- Directories to check  (perform all possible verifications) -->
  <directories check_all="yes" realtime="yes">/etc</directories>
  <directories check_all="yes" realtime="yes">/home</directories>
```

We changed the frequency of FIM runs every **43200 seconds = 12 HOURS** to **60 seconds**

Due to this changes are not being detected immediately

We enabled real-time file integrity monitoring on critical directories like /etc and user directories like /home using Wazuh syscheck module with full attribute verification.

RESULTS:

Mar 22, 2026 @ 06:10:22.784	kali-client	Integrity checksum changed.	7	550
Mar 22, 2026 @ 06:10:22.779	kali-client	Integrity checksum changed.	7	550
Mar 22, 2026 @ 06:10:17.480	kali-client	Integrity checksum changed.	7	550
Mar 22, 2026 @ 06:10:16.445	kali-client	Integrity checksum changed.	7	550
Mar 22, 2026 @ 06:10:16.414	kali-client	Integrity checksum changed.	7	550
Mar 22, 2026 @ 06:10:16.359	kali-client	File deleted.	7	553
Mar 22, 2026 @ 06:10:16.349	kali-client	Integrity checksum changed.	7	550
Mar 22, 2026 @ 06:10:16.342	kali-client	Integrity checksum changed.	7	550
Mar 22, 2026 @ 06:10:16.335	kali-client	File added to the system.	5	554
Mar 22, 2026 @ 06:10:16.232	kali-client	Integrity checksum changed.	7	550
Mar 22, 2026 @ 06:10:15.438	kali-client	Integrity checksum changed.	7	550
Mar 22, 2026 @ 06:10:14.203	kali-client	Integrity checksum changed.	7	550
Mar 22, 2026 @ 06:10:14.200	kali-client	File deleted.	7	553
Mar 22, 2026 @ 06:10:14.197	kali-client	Integrity checksum changed.	7	550
Mar 22, 2026 @ 06:10:14.180	kali-client	File added to the system.	5	554
Mar 22, 2026 @ 06:10:13.522	kali-client	Integrity checksum changed.	7	550
Mar 22, 2026 @ 06:10:13.509	kali-client	Integrity checksum changed.	7	550

Conclusion

File Integrity Monitoring was successfully configured by enabling real-time monitoring on key directories such as /etc and /home. Upon modifying test files, Wazuh generated alerts including file addition, deletion, and checksum changes. This demonstrates the system's ability to detect unauthorized changes instantly, enhancing system security and integrity monitoring.

Week 3

Objective

To implement **Active Response mechanisms** in Wazuh for automatically detecting and mitigating brute-force attacks by monitoring SSH authentication failures and triggering predefined security rules.

Setting up the rules for global configuration

Before:

```
kali@kali: ~
Session  Actions  Edit  View  Help
  GNU nano 8.7.1                                  /var/ossec/etc/ossec.conf *

  <localfile>
    <log_format>full_command</log_format>
    <command>netstat -tulpn | sed 's/\([[:alnum:]]\+\)\ \+[[:digit:]]\+\ \+[[:digit:]]\+\ \+\(.*\):\([[:digit:]]*\)\ \+\([0-9\.\:\*]\+\).\+\ \([[:digit:]]*\>
    <alias>netstat listening ports</alias>
    <frequency>360</frequency>
  </localfile>

  <localfile>
    <log_format>full_command</log_format>
    <command>last -n 20</command>
    <frequency>360</frequency>
  </localfile>

  <!-- Active response -->
  <active-response>
    <disabled>no</disabled>
    <ca_store>etc/wpk_root.pem</ca_store>
    <ca_verification>yes</ca_verification>
  </active-response>

  <!-- Choose between "plain", "json", or "plain,json" for the format of internal logs -->
  <logging>
    <log_format>plain</log_format>
  </logging>

</ossec_config>
```

After:

```
kali@kali: ~
Session  Actions  Edit  View  Help
  GNU nano 8.7.1                                  /var/ossec/etc/ossec.conf *

  <localfile>
    <log_format>full_command</log_format>
    <command>netstat -tulpn | sed 's/\([[:alnum:]]\+\)\ \+[[:digit:]]\+\ \+[[:digit:]]\+\ \+\(.*\):\([[:digit:]]*\)\ \+\([0-9\.\:\*]\+\).\+\ \([[:digit:]]*\>
    <alias>netstat listening ports</alias>
    <frequency>360</frequency>
  </localfile>

  <localfile>
    <log_format>full_command</log_format>
    <command>last -n 20</command>
    <frequency>360</frequency>
  </localfile>

  <!-- Active response -->
  <active-response>
    <disabled>no</disabled>
    <ca_store>etc/wpk_root.pem</ca_store>
    <ca_verification>yes</ca_verification>
    <command>firewall-drop</command>
    <location>local</location>
    <rules_id>5710</rules_id>
    <timeout>60</timeout>
  </active-response>

  <!-- Choose between "plain", "json", or "plain,json" for the format of internal logs -->
  <logging>
    <log_format>plain</log_format>

^G Help      ^O Write Out   ^F Where Is    ^K Cut       ^T Execute   ^C Location  M-U Undo      M-A Set Mark   M-] To Bracket
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo    M-6 Copy       ^B Where Was
```

Line	Meaning
firewall-drop	Block attacker IP
location=local	Block on Kali machine

Line	Meaning
rules_id=5710	Trigger on brute force
timeout=60	Block for 60 seconds

Result:

Mar 22, 2026 @ 07:30:10.741	kali-client	syslog: User missed the password more than one time	10	2502
Mar 22, 2026 @ 07:30:08.736	kali-client	sshd: Attempt to login using a non-existent user	5	5710
Mar 22, 2026 @ 07:30:04.725	kali-client	sshd: Attempt to login using a non-existent user	5	5710
Mar 22, 2026 @ 07:30:03.001	kali-client	Integrity checksum changed.	7	550
Mar 22, 2026 @ 07:30:02.986	kali-client	Integrity checksum changed.	7	550
Mar 22, 2026 @ 07:30:00.715	kali-client	sshd: Attempt to login using a non-existent user	5	5710
Mar 22, 2026 @ 07:29:58.711	kali-client	PAM: User login failed.	5	5503
Mar 22, 2026 @ 07:29:54.703	kali-client	sshd: Attempt to login using a non-existent user	5	5710

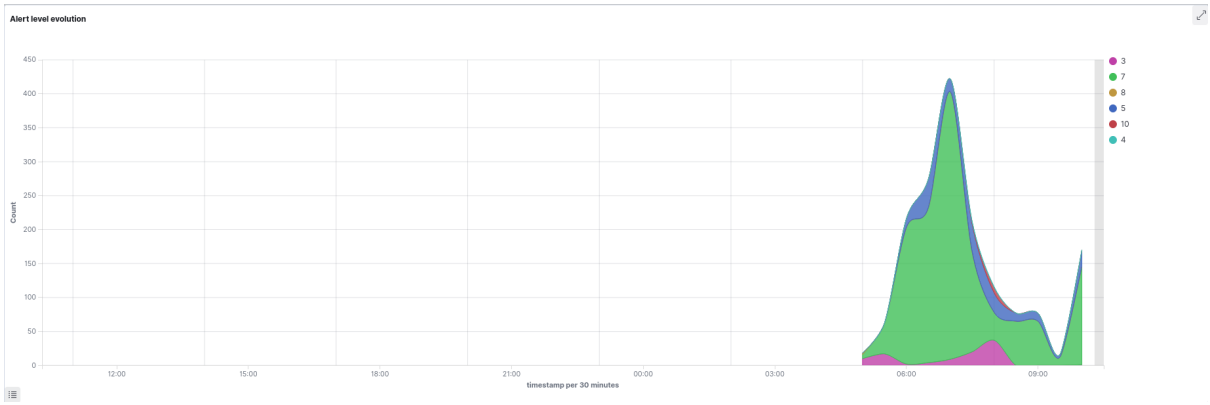
Conclusion

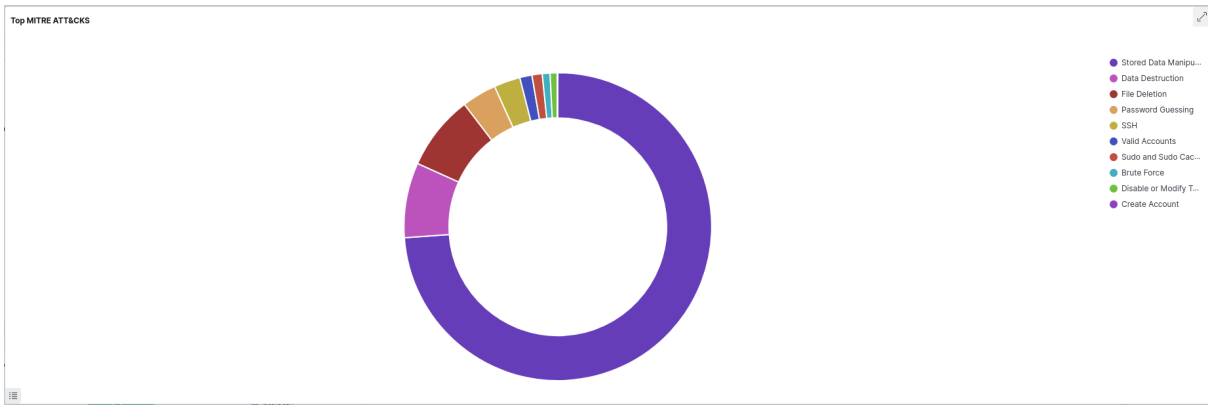
SSH brute-force attack simulation was successfully performed using multiple failed login attempts. Wazuh detected these attempts through rule-based analysis and generated alerts such as “authentication failure” and “non-existent user login attempts.” The system was configured to initiate active response actions, enabling automatic mitigation of potential threats, thereby enhancing real-time security defense capabilities.

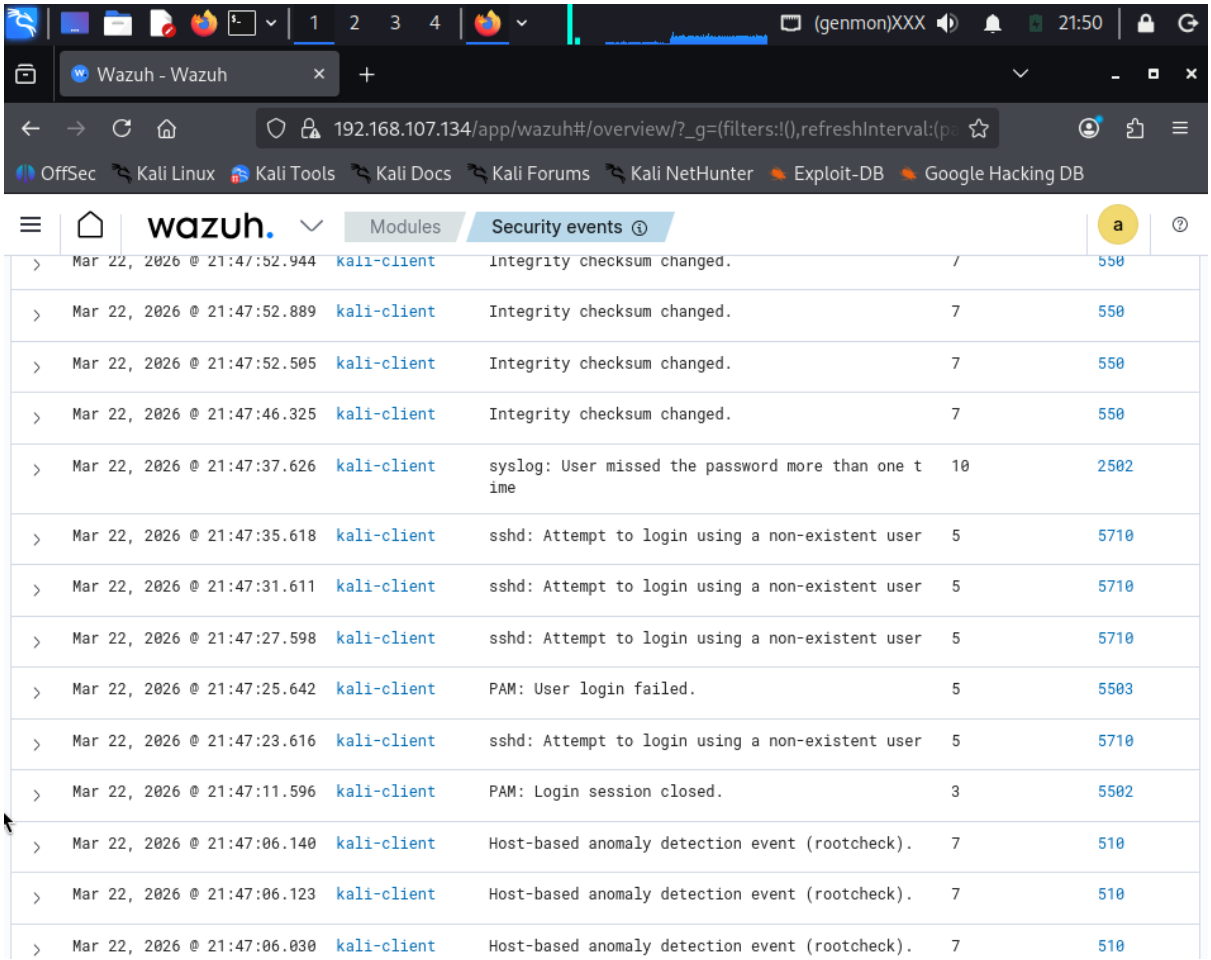
Week 4

Objective

To simulate various security threats and analyze their detection using Wazuh dashboard visualization tools, enabling real-time monitoring and security event analysis.







Mar 22, 2026 @ 21:47:52.944	kali-client	Integrity checksum changed.	7	550
Mar 22, 2026 @ 21:47:52.889	kali-client	Integrity checksum changed.	7	550
Mar 22, 2026 @ 21:47:52.505	kali-client	Integrity checksum changed.	7	550
Mar 22, 2026 @ 21:47:46.325	kali-client	Integrity checksum changed.	7	550
Mar 22, 2026 @ 21:47:37.626	kali-client	syslog: User missed the password more than one time	10	2502
Mar 22, 2026 @ 21:47:35.618	kali-client	sshd: Attempt to login using a non-existent user	5	5710
Mar 22, 2026 @ 21:47:31.611	kali-client	sshd: Attempt to login using a non-existent user	5	5710
Mar 22, 2026 @ 21:47:27.598	kali-client	sshd: Attempt to login using a non-existent user	5	5710
Mar 22, 2026 @ 21:47:25.642	kali-client	PAM: User login failed.	5	5503
Mar 22, 2026 @ 21:47:23.616	kali-client	sshd: Attempt to login using a non-existent user	5	5710
Mar 22, 2026 @ 21:47:11.596	kali-client	PAM: Login session closed.	3	5502
Mar 22, 2026 @ 21:47:06.140	kali-client	Host-based anomaly detection event (rootcheck).	7	510
Mar 22, 2026 @ 21:47:06.123	kali-client	Host-based anomaly detection event (rootcheck).	7	510
Mar 22, 2026 @ 21:47:06.030	kali-client	Host-based anomaly detection event (rootcheck).	7	510

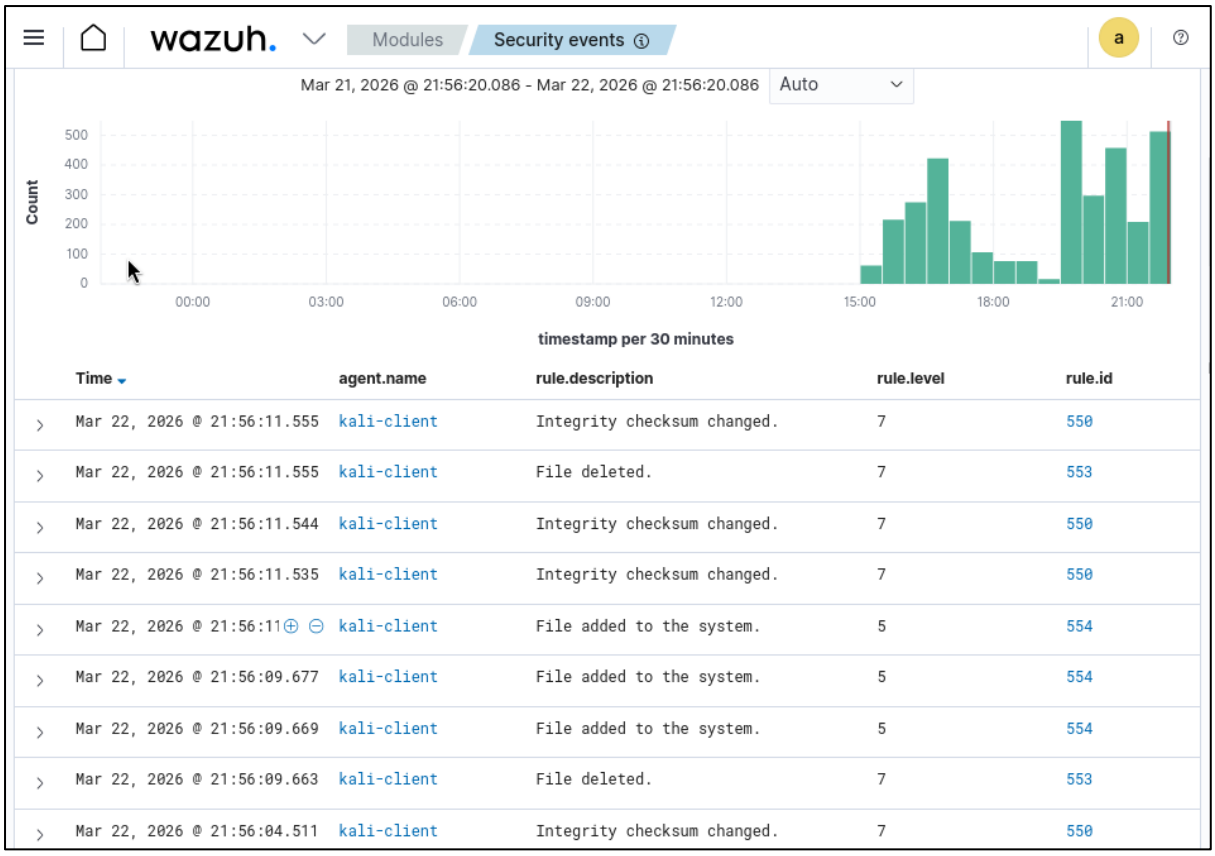
Simulating “Ransomware-like” Attack (T1490 style)

```
┌──(kali㉿kali)-[~]
└─$ # Simulate deletion of important data
rm -rf /home/kali/important_file.txt

# Simulate encryption behavior
echo "ENCRYPTED_DATA" > /home/kali/important_file.txt

# Change permissions (suspicious behavior)
chmod 000 /home/kali/important_file.txt
```

What this simulates

Action	**Real Attack**
Delete file	Data destruction
Rewrite file	Encryption
Permission change	Locking victim out

Result:



Time	agent.name	rule.description	rule.level	rule.id
Mar 22, 2026 @ 21:56:11.555	kali-client	Integrity checksum changed.	7	550
Mar 22, 2026 @ 21:56:11.555	kali-client	File deleted.	7	553
Mar 22, 2026 @ 21:56:11.544	kali-client	Integrity checksum changed.	7	550
Mar 22, 2026 @ 21:56:11.535	kali-client	Integrity checksum changed.	7	550
Mar 22, 2026 @ 21:56:11	kali-client	File added to the system.	5	554
Mar 22, 2026 @ 21:56:09.677	kali-client	File added to the system.	5	554
Mar 22, 2026 @ 21:56:09.669	kali-client	File added to the system.	5	554
Mar 22, 2026 @ 21:56:09.663	kali-client	File deleted.	7	553
Mar 22, 2026 @ 21:56:04.511	kali-client	Integrity checksum changed.	7	550

Kill chain observed

1. **Initial Access:**
   SSH brute-force attack simulated
2. **Execution:**
   Unauthorized commands executed on system
3. **Persistence:**
   File creation and modification observed
4. **Impact:**
   File deletion and simulated encryption (ransomware behavior)
5. **Detection:**
   Wazuh generated alerts for file changes and login failures

Conclusion

A ransomware-like attack was successfully simulated by performing file deletion, modification, and permission changes. Wazuh detected these activities and generated alerts corresponding to impact-based MITRE ATT&CK techniques. The alerts were visualized in the dashboard, and the attack sequence was analyzed using the kill chain model, demonstrating effective threat detection and monitoring capabilities.